Antiderivatives and Indefinite Integrals

ANSWER KEY

Section 1: The Power Rule for Integration

1 Evaluate $\int (6x^3 - 4x + 5)\,dx$.

a. $\frac{3}{2}x^4 - 2x^2 + 5x + C$ ✓

b. $2x^4 - 2x^2 + 5x + C$

c. $18x^2 - 4 + C$

d. $\frac{3}{2}x^4 - 4x^2 + 5x + C$

$\int 6x^3\,dx = 6(x^4/4) = \frac{3}{2}x^4$. $\int -4x\,dx = -2x^2$. $\int 5\,dx = 5x$. Result: $\frac{3}{2}x^4 - 2x^2 + 5x + C$.

2 Find $\int \left(x^{2/3} + \frac{1}{x^3}\right) dx$.

a. $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$ ✓

b. $\frac{5}{3}x^{5/3} - \frac{1}{4x^4} + C$

c. $\frac{3}{5}x^{5/3} + \frac{1}{2x^2} + C$

d. $\frac{2}{5}x^{5/3} - \frac{2}{x^2} + C$

$\int x^{2/3}\,dx = \frac{x^{5/3}}{5/3} = \frac{3}{5}x^{5/3}$. $\int x^{-3}\,dx = \frac{x^{-2}}{-2} = -\frac{1}{2x^2}$. Result: $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$.

3 Evaluate $\int \frac{3x^4 - 2x}{x^2}\,dx$.

a. $\frac{3}{3}x^3 - 2\ln|x| + C = x^3 - 2\ln|x| + C$ ✓

b. $x^3 - \frac{2}{x} + C$

c. $3x^3 - 2x + C$

d. $x^3 - 2x^{-1} + C$

Simplify integrand: $\frac{3x^4}{x^2} - \frac{2x}{x^2} = 3x^2 - \frac{2}{x}$. Integrating: $x^3 - 2\ln|x| + C$.

Section 2: Trigonometric and Exponential Antiderivatives

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4 Evaluate $\int (3\sin x + 4\cos x)\,dx$.

a. $-3\cos x + 4\sin x + C$ ✓

b. $3\cos x - 4\sin x + C$

c. $3\cos x + 4\sin x + C$

d. $-3\cos x - 4\sin x + C$

$\int \sin x\,dx = -\cos x$ and $\int \cos x\,dx = \sin x$. Thus $\int (3\sin x + 4\cos x)\,dx = -3\cos x + 4\sin x + C$.

5 Evaluate $\int (5e^x + \sec^2 x)\,dx$.

a. $5e^x + \tan x + C$ ✓

b. $5e^x - \tan x + C$

c. $5e^x + \sec x \tan x + C$

d. $e^x + \tan x + C$

$\int e^x\,dx = e^x$ and $\int \sec^2 x\,dx = \tan x$. Result is $5e^x + \tan x + C$.

6 Find $\int (2^x + \csc x \cot x)\,dx$.

a. $\frac{2^x}{\ln 2} - \csc x + C$ ✓

b. $2^x \ln 2 - \csc x + C$

c. $\frac{2^x}{\ln 2} + \csc x + C$

d. $2^x - \cot x + C$

$\int a^x\,dx = \frac{a^x}{\ln a}$ and $\int \csc x \cot x\,dx = -\csc x$. Result is $\frac{2^x}{\ln 2} - \csc x + C$.

Section 3: Initial Value Problems

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7 Find $f(x)$ given $f'(x) = 4x - 3$ and $f(1) = 6$.

a. $f(x) = 2x^2 - 3x + 7$ ✓

b. $f(x) = 2x^2 - 3x + 5$

c. $f(x) = 4x^2 - 3x + 5$

d. $f(x) = 2x^2 - 3x$

$f(x) = \int (4x - 3)\,dx = 2x^2 - 3x + C$. $f(1) = 2(1)^2 - 3(1) + C = 6 \Rightarrow -1 + C = 6 \Rightarrow C = 7$.

8 A curve satisfies $\frac{dy}{dx} = 3x^2 + 2x$ and passes through $(2, 10)$. Find its equation.

a. $y = x^3 + x^2 - 2$ ✓

b. $y = x^3 + x^2 + 2$

c. $y = 3x^3 + 2x^2 - 22$

d. $y = x^3 + x^2$

$y = x^3 + x^2 + C$. $y(2) = 8 + 4 + C = 10 \Rightarrow 12 + C = 10 \Rightarrow C = -2$. Equation: $y = x^3 + x^2 - 2$.

9 Given $f''(x) = 6x$, $f'(0) = 4$, and $f(0) = 1$, find $f(x)$.

a. $f(x) = x^3 + 4x + 1$ ✓

b. $f(x) = 3x^3 + 4x + 1$

c. $f(x) = x^3 + 2x + 1$

d. $f(x) = x^3 + 4$

$f'(x) = 3x^2 + C_1$. $f'(0) = 4 \Rightarrow C_1 = 4$. $f(x) = x^3 + 4x + C_2$. $f(0) = 1 \Rightarrow C_2 = 1$.

Section 4: Rectilinear Motion and Acceleration

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10 A particle moves with acceleration $a(t) = 6t - 4$. If its initial velocity $v(0) = 5$, find $v(t)$.

a. $v(t) = 3t^2 - 4t + 5$ ✓

b. $v(t) = 6t^2 - 4t + 5$

c. $v(t) = 3t^2 - 4t$

d. $v(t) = 3t^2 + 4t + 5$

$v(t) = \int (6t - 4)\,dx = 3t^2 - 4t + C$. $v(0) = 5 \Rightarrow C = 5$. Thus $v(t) = 3t^2 - 4t + 5$.

11 If a ball is thrown upward with $v(0) = 48$ ft/s and $a(t) = -32$ ft/s^2 from ground level $s(0) = 0$, find its maximum height.

a. 36 ft ✓

b. 48 ft

c. 64 ft

d. 32 ft

$v(t) = -32t + 48 = 0 \Rightarrow t = 1.5$ s. $s(t) = -16t^2 + 48t$. At $t = 1.5$:
$s(1.5) = -16(2.25) + 48(1.5) = -36 + 72 = 36$ ft.

12 A car decelerates at a constant $a = -4$ m/s^2 from $v_0 = 24$ m/s. How far does it travel before coming to a stop?

a. 72 m ✓

b. 48 m

c. 96 m

d. 36 m

$v(t) = 24 - 4t = 0 \Rightarrow t = 6$ s. Distance $s(t) = 24t - 2t^2$. At $t = 6$:
$s(6) = 24(6) - 2(36) = 144 - 72 = 72$ m.

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